

LLM-Based User Personas for Recommendations at Scale



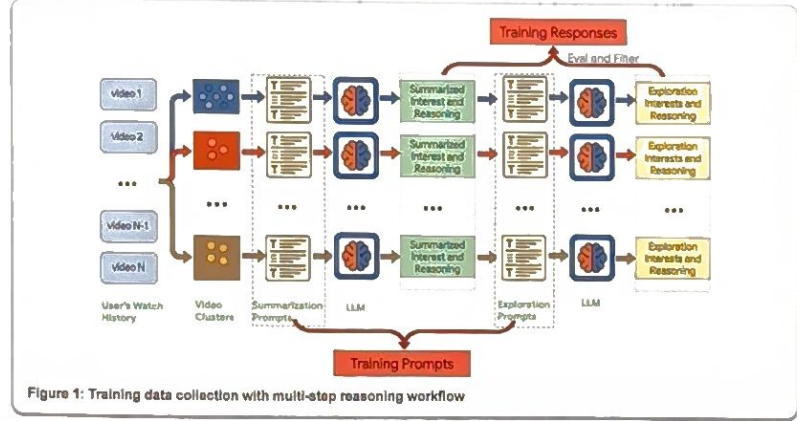
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Motivation

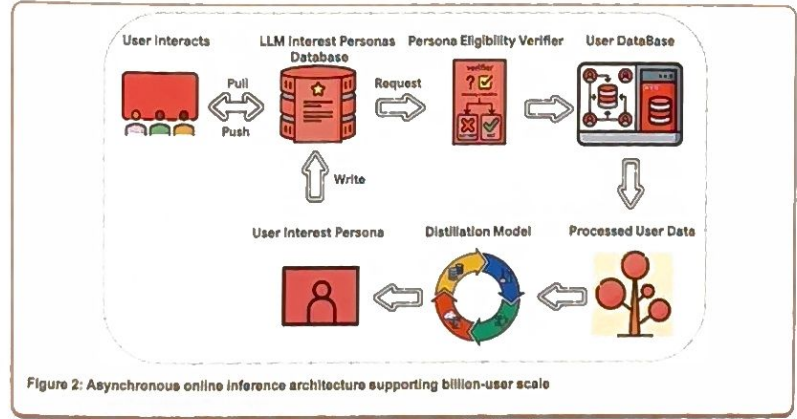
- **Beyond IDs:** Traditional ID embeddings capture co-occurrence but remain opaque, lacking the semantic context, world knowledge, and human interpretability needed for user-facing features.
- **Real-Time Serving vs. Offline Lookups:** Prior LLM recommenders pre-compute static lookups over small fixed vocabularies, missing immediate user intent and rapidly evolving interests.
- **Dual-Objective Natural-Language Personas:** A single unified inference call synthesizes holistic user personas balancing
 - **Summarized Interests (Exploitation):** Concise, human-readable textual labels synthesizing existing user preferences from watch history.
 - **Exploration Interests (Discovery):** Novel yet related topics inferred via LLM world knowledge to uncover fresh user affinities.

Overview / Methodology

- **Multi-Step CoT Distillation (Figure 1):** Off-the-shelf small models fail on complex dual tasks. We distill Gemini Pro into Gemini Nano
- **Infinite Concept Clustering:** Groups watch history into semantic clusters via salient-term cosine similarity in [0, 1] and prunes sparse clusters
- **Decomposed CoT Teacher:** Separately generates & filters Summarization and Exploration reasoning traces (80% Train / 20% Eval split).



- **Asynchronous Online Serving (Figure 2):** Decouples LLM generation from live requests for minimal serving latency.
- **Async Refresh & Safety Fallback:** Serves cached personas; background job regenerates stale entries via distilled Gemini Nano + Safety Classifier fallback
- **Hybrid Hierarchical Retrieval:** Samples from Summarized and Exploration interests per request to guide sequential transformer via restricted nearest neighbor search



- **Prompt Design: Structured Input Grouping & Unified Dual-Task Serving**

Prompt B.1: Unified Summarization + Exploration

I'm a brilliant video topic summarization expert who speaks all languages. Given a few groups of videos a person watched, I can complete the following 2 tasks:

Task 1: Describe the interests of that person for each group and explain why, respectively. I can wrap the interests of that person using "".

Task 2: For each summarized interest in Task 1, think of 3 creative and exploratory interests that are relevant to that summarized interest, but also novel and provide new perspectives. I can wrap the exploration interests in this task using &&.

For example: ...

As another example: ...

Now, a person watched following <Num_Groups> groups of videos:

{Group 0}: video_metadata1, ..., video_metadataM0

{Group 1}: video_metadata1, ..., video_metadataM1

...

My output is: ...

Dual-Objective CoT Serving

User Survey: Satisfaction On LLM-Generated Interests

- **>80% Perceived Accuracy:** Over 80% of respondents reported the LLM-generated label summarized their watch history "Very Closely" or "Extremely Closely"
- **71% Strong Viewing Preference:** 71% of users expressed strong interest in watching more videos related to the LLM-generated topic
- **Preferred Over Knowledge Graph Baseline:** 57% of users strictly preferred LLM personas over Knowledge Graph entity topics; an additional 20% rated them equivalent
- **Main Reasons for User Dissatisfaction:** (1) Missing primary interests, (2) inferring interests from sporadic activity, (3) outdated interests, and (4) repetitive labels

Offline Study: User Representation & Distillation

- **Semantic Clustering:** Grouping watch history into semantic clusters significantly outperforms chronological order and audiovisual clusters
- **Video Titles:** Titles outperform noisy descriptions (promotional text/credits) and overly broad salient terms
- **Model Size:** Gemini Pro and Ultra outperform Flash, plateau from Pro to Ultra motivates distilling large teacher models into efficient students

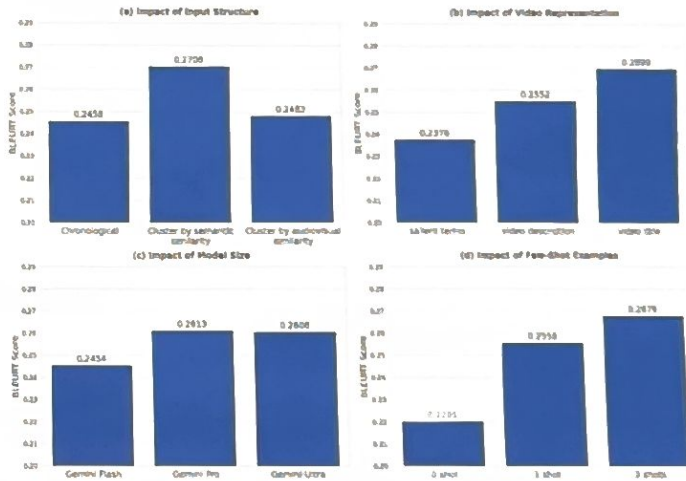


Figure 3: Offline BLEURT score across (a) Input Structure, (b) Video Metadata, (c) Model Size, and (d) Few-Shot Prompting

Table 1: Qualitative Comparison of Summarized Interests

Input Structure	LLM Output
Sequential (Prompt A.1)	bengali tv shows
Semantic Clustered (Clustered Input)	bengali tv series "mon ditey chal" bengali tv series "icche putul" & "kar kache kol moner kotha" bengali tv series "bojhena se bojhena"

Table 2: Student Model Distillation: Distilled Gemini Nano lifts Instruction Following Rate (IFR) from 0.07% → 99.08% and achieves BLEURT (0.328) comparable to larger models at low serving cost

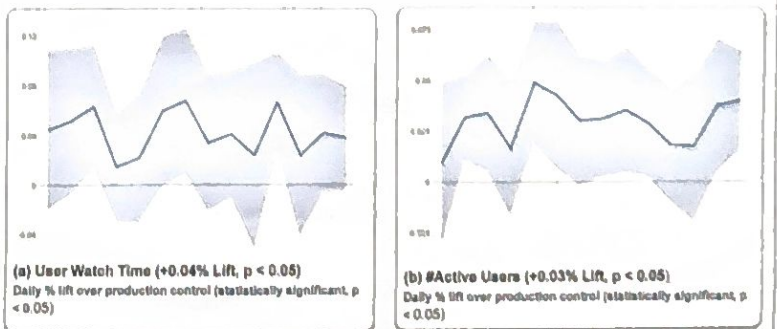
Note: (-) checkpoints not evaluated. BLEURT & Creativity computed on instruction-compliant samples. N/A indicates insufficient valid samples due to near-zero IFR at epoch 0

Epoch	IFR (%) Nano	IFR (%) Flash	BLEURT Nano	BLEURT Flash	Creativity Nano	Creativity Flash
0.00	0.07	1.82	N/A	0.286	N/A	-0.588
4.37	-	99.54	-	0.334	-	0.008
10.82	98.97	99.76	0.321	0.338	-0.012	0.018
26.20	99.08	99.68	0.328	0.345	-0.018	0.023

Live Experiment (30+ Days, Billion-User Scale)

- **Statistically Significant Viewer Value Lift (Figure 4):** +0.04% User Watch Time ($p < 0.05$) & +0.03% Active Users ($p < 0.05$) over 30+ days at billion-user scale, driven primarily by casual users.
- **Sustained Topic Exploration:** +0.04% Engaged Topics ($p < 0.05$) & +0.03% users with multiple lasting engaged topics ($p < 0.05$), converting serendipitous discovery into repeat visits.
- **Exposure Gap:** Exploration items receive 40.91% fewer impressions than summarized items due to downstream ranking model bias.
- **Recommendation Efficiency:** Conditional on presentation, exploration items are 13.6% more likely to be watched than summarized items, proving high user affinity.

Figure 4: 30+ day live experiment trajectory (Y-axis: % lift over production control, X-axis: time).



Key Takeaways & Production Impact

- **Dual-Task LLM Personas:** Jointly extracting core summarized interests and exploratory expansions generates rich, interpretable semantic profiles without separate pipelines
- **Distillation for Production Serving:** Distillation lifts instruction-following to 99.08% while satisfying millisecond latency at low serving cost.
- **Large-Scale User Impact:** Validated in 30+ day live YouTube experiments (+0.04% Watch Time, +0.03% Active Users, $p < 0.05$) and >80% user-perceived accuracy in user studies